

When conditional variance harms KSIV on the X-shaped target

Geometry, representation noise, and optimization boundaries

53 new remote runs; three-seed long comparisons; 334 historical checkpoints audited.

The main conclusion

The gap is a failure mode of finite stochastic optimization, not an intrinsic inferiority of the conditional variational family. The larger family contains the global family, and a fixed-kernel population KSD depends only on the marginal distribution. For this target, a global variance of 0.2 already supplies exactly the needed transverse width. Conditional scale directions can instead develop small-variance tails that sharply increase conditional-score gradient noise. The optimizer and moving target determine whether that noise leads to failure.

The evidence is deliberately qualified: not every conditional run fails, not every global run succeeds, and the intervention results do not imply a universal preference for the Stein estimator. The full standalone manuscript, `report.tex`, supplies additional derivations, per-seed diagnostics, and evaluation details.

The exact comparison

Canonical policy: two width-128 SiLU hidden layers, Gaussian latent dimension 2, softplus conditional variances with floor 0.0001; Adam (0.9, 0.999), learning rate 0.001, StepLR(1000, 0.9), two independent batches of 128, and 50,000 updates. The target-score multiplier moves linearly from 0.1 to 1 over 25,000 updates. Gaussian spatial kernels and the empirical median bandwidth are differentiated. New comparisons match mean weights and initial variance to float32 roundoff and pair the training noise. Diagnostics preserve training RNG.

Long comparisons: means (sample standard deviations across trained seeds). SW2 is sliced Wasserstein distance; KL is forward KL estimated from exact target samples. Independent KSD uses a fixed $h=0.75$. Each trained model, not each evaluation sample, is one statistical replicate.

Procedure	n	SW2	KL(p q)	KSD squared
Conditional: canonical	3	1.965 (1.136)	1.161 (0.378)	0.0449 (0.0307)
Global: canonical	3	0.214 (0.130)	0.275 (0.233)	0.0025 (0.0018)
Conditional: Stein estimator	3	0.283 (0.172)	0.357 (0.298)	0.0038 (0.0011)
Global: Stein estimator	3	0.371 (0.018)	0.874 (0.182)	0.0039 (0.0009)
Conditional: variance floor 0.2	3	0.400 (0.024)	0.579 (0.029)	0.0057 (0.0015)
Conditional: no anneal, constant LR	3	1.276 (1.486)	0.600 (0.596)	0.0528 (0.0770)
Global: no anneal, constant LR	3	0.167 (0.022)	0.049 (0.022)	0.0013 (0.0008)
Conditional: detach bandwidth	3	1.360 (1.050)	1.004 (0.420)	0.0193 (0.0115)
Conditional: fixed $h=0.75$	3	0.589 (0.168)	0.693 (0.277)	0.0061 (0.0016)
Conditional: fixed h , Stein	3	0.565 (0.188)	0.684 (0.220)	0.0104 (0.0067)

The canonical gap occurs in all three paired seeds: mean SW2 is 1.965 versus 0.214, and forward KL is 1.161 versus 0.275. Raising the conditional variance floor to 0.2 improves every canonical conditional seed, with mean SW2 0.400. The adaptive-bandwidth Stein conditional variant also improves every canonical conditional

seed and has mean SW2 0.283; detaching bandwidth alone leaves mean SW2 1.360.

Theory 1: population representation invariance

$$f = s_p(X) - s_c(X, \varepsilon) = s_p(X) + U/\sigma(\varepsilon)$$

$$\mathbb{E}[s_c(X, \varepsilon) \mid X] = s_q(X)$$

For two independent hierarchical draws, conditional expectation converts $\mathbb{E}[k(X, X') f(X, \varepsilon) f(X', \varepsilon)]$ into the marginal KSD squared. The same marginal q therefore has the same population loss under different Gaussian-mixture representations. Constant variance is available inside the conditional network by zeroing variance-output weights, so its globally optimized fixed-kernel loss cannot be worse. This says nothing about finite Adam optimization or a data-adaptive kernel.

Theory 2: the exact global-variance boundary

The target is an equal mixture of centered Gaussians with diagonal entries 2 and off-diagonal entries +1.8 or -1.8. Its component eigenvalues are 3.8 and 0.2. Let S be an equiprobable sign and T and U be independent standard normal draws:

$$X = \sqrt{1.8} T(1, S)^T + \sqrt{0.2} U$$

This has exactly the target distribution: mixing means supply the long arms and global noise supplies transverse width. A finite continuous neural mean approximates the branch switch; the construction is not a claim of exact representation by a finite width-128 network.

$$D = \text{diag}(d_1, d_2) : (2 - d_1)(2 - d_2) \geq 3.24, \quad d_1, d_2 \leq 2$$

Proof of the boundary. These inequalities are equivalent to both residual covariance matrices being positive semidefinite, which suffices by Gaussian convolution. If a residual covariance is negative in a direction, dividing the target characteristic function by the noise characteristic function gives a positive exponential term growing without bound. A characteristic function has absolute value at most one, so exact deconvolution is impossible. For isotropic $D=vI$ the exact boundary is $0 \leq v \leq 0.2$. For off-diagonals $\pm 2\rho$ it becomes $v \leq 2(1-|\rho|)$.

Theory 3: latent representation noise at an exact optimum

$$q = p : \mathbb{E}\|s_p - s_c\|^2 = \mathbb{E} \text{tr}(D^{-1}) - \mathbb{E}_p\|s_p\|^2$$

Proof. Expand the squared residual and use $\mathbb{E}[s_c|X] = s_p(X)$. A Gaussian conditional score has second moment $\text{tr}(D^{-1})$. Thus the residual can be noisy even when the marginal is exactly correct. Jensen gives $\mathbb{E}[1/v] \geq 1/\mathbb{E}[v]$. A moderate average variance can conceal substantial noise from a small-variance tail. This identity concerns score noise; it does not alone prove an increase in every gradient coordinate.

Theory 4: an analytic gradient-noise explosion

For any $0 < v \leq 0.2$, use mixing means from the residual covariance mixture and add independent $N(0, vI)$ noise. The marginal remains exactly p for every v . Differentiate a global dilation $X(\theta) = \exp(\theta)X$, $\sigma(\theta) = \exp(\theta)\sigma$ at $\theta=0$, using an independently fixed Gaussian bandwidth h :

$$\lim_{v \rightarrow 0} v^2 \text{Var}(g_v) = \frac{2}{N^2} \mathbb{E}_{p \otimes p} \left[k_h^2 \left(2 + \frac{\|X - X'\|^2}{h^2} \right)^2 \right] > 0$$

Proof. The leading pair-gradient term is $-k(X, X')[2 + \|X - X'\|^2/h^2] U^T U'/v$. Couple the residual Gaussian means so they converge to independent target draws as v approaches zero. Polynomial growth of the target score derivatives and Gaussian moments give L2 convergence of v times the gradient. Different pair summands have zero covariance even if they share one index, because the other Gaussian noise is centered. $E[(U^T U')^2] = 2$, giving the positive limit. The actual variance floor is positive; this is a mechanism for large finite variance constants, not a claim of infinite runtime variance.

The Stein integration-by-parts gradient is a deterministic function of the independent marginal samples X and X' . Its full dilation-gradient distribution is therefore exactly representation invariant in this construction. Dilating only mean outputs still gives inverse-square conditional-gradient variance growth, with the squared bracket replaced by $\|X - X'\|^4/h^4$; the Stein mean-direction second moment stays bounded. Scaling final-layer mean rows is an available network parameter direction, so this mechanism can contaminate mean updates too.

The variance constant is computed from Gaussian quadratic-form Laplace transforms. $X - X'$ has covariance eigenvalues (7.6, 0.4) or (4, 4), equally weighted. At $h=0.75$ and $N=128$, the predicted asymptotic variance is $0.000099824/v^2$. No parameter is fitted to the gradient experiment.

Component variance	Measured conditional variance	Measured Stein variance	Analytic asymptotic
0.2	0.00253	0.00196	0.00250
0.11	0.01002	0.00201	0.00825
0.05	0.04286	0.00204	0.03993
0.02	0.28042	0.00217	0.24956
0.02 / 0.20 mixture	0.07267	0.00186	-
0.002	24.45727	0.00203	24.95605

Each row uses 256 independent repetitions, two batches of 128, and exactly the same marginal p . At $v=0.002$ the prediction is 24.96 and measurement 24.46, while the Stein variance is about 0.002. Randomizing variance between 0.02 and 0.20 keeps both marginal p and mean variance 0.11, but raises the measured conditional gradient variance about sevenfold relative to constant 0.11. This is a distributional counterfactual; architecture-specific causal evidence comes from training controls.

What the training controls do and do not establish

The Stein training variant removes explicit U/σ from the integrand. The bandwidth-only control also detaches the median but retains spatial kernel gradients. Their comparison separates bandwidth detachment from the estimator change. A second three-seed contrast fixes $h=0.75$ in both estimators, so it also removes adaptive-bandwidth bias. The population-equivalence claim is restricted to fixed independent kernels: jointly data-dependent bandwidths introduce additional integration-by-parts terms.

The variance-floor intervention bounds conditional variance below by 0.2; the exact target construction remains available in the flexible-mean closure. It also changes gradient geometry and introduces clamped zero gradients, so it is not a pure noise intervention. Removing annealing and sustaining learning rate does not guarantee rescue: outcomes differ across seeds, and at least one conditional run under that policy beats its global counterpart.

The independently fixed-bandwidth training contrast sets an important limit: Stein improves SW2 for seeds 42 and 43 but worsens it for seed 44. Its mean forward KL is nearly unchanged and mean held-out KSD is worse. Thus eliminating explicit inverse-variance terms does not ensure better optimization; the large adaptive-bandwidth improvement cannot be transferred to every kernel policy. Conversely, a successful Stein conditional run still has a large inverse-variance tail. Small variance can represent the target well; the

stochastic estimator determines whether that representation is difficult to optimize.

Broad screening varied batch size, floors, shared versus separate networks, annealing duration, decay, estimator, bandwidth differentiation, and correlation. These one-seed runs lasted 10,000 updates; annealed runs still target a tempered density at that horizon. The isotropic Gaussian control shows no comparable conditional disadvantage. Shared-trunk removal or larger batches alone do not establish a full explanation. An exact empirical critical correlation is not located by these tests.

Why the convergence theorem does not settle the observation

Annealing reaches the final target after learning rate has fallen to 0.0000718. Only about 6.7% of total nominal step-size mass is spent on that final target. This is not an Adam-displacement bound, but explains limited late adaptation. The target also has unbounded Hessian:

$$\log p(x, y) = \text{const} - \frac{a}{2}(x^2 + y^2) + \log \cosh(bxy), \quad a = 2/0.76, \quad b = 1.8/0.76$$
$$\partial_{xx} \log p(0, R) = -a + b^2 R^2 \rightarrow \infty$$

The global bounded-Hessian assumption of the published optimization theorem is therefore not literally met by this toy target. This applies to both families and is not itself a cause of their ordering. The theorem also concerns stationarity under stated assumptions, not global minimization or small KL. The observation does not contradict a guarantee of family expressiveness or successful global optimization.

Evidence, reproducibility, and limits

All runs used the specified server after reloading AGENTS.md (port 37874), an RTX 4090, PyTorch 2.9.0+cu126, and the ruivi environment. Jobs ran in tmux; code and generated reports were synchronized through Git. Raw logs, samples, checkpoints, and state are under /root/ruivi/results/ksivi_variance_investigation_20261006. Each run records its own code revision. The fast research objective gradient was verified against the production update on CPU and GPU.

Held-out SW2 uses 10,000 samples and 128 directions. KL uses 4,096 exact target draws and nested explicit Gaussian mixtures; integration increases to 262,144 latent draws when needed. Finite latent integration remains a source of KL uncertainty. Independent KSD uses 16 pairs of 256 samples and fixed bandwidths 0.5, 0.75, and 1.5. Tables and JSON retain every seed, evaluation standard errors, latent integration checks, and failed interventions. Covariance alone is insufficient: target cross-fourth moment in rotated coordinates is 0.76, compared with 4 for a covariance-matched isotropic Gaussian.

The analytical identities, exact representation boundary, and gradient asymptotic are proved. The training comparisons and unchanged-marginal simulation are measured. The combined account of stochastic noise, geometry, and basin selection is a supported interpretation, not a proof that one scalar diagnostic determines every outcome. Three seeds do not establish universal optimizer, kernel, or target boundaries.

References

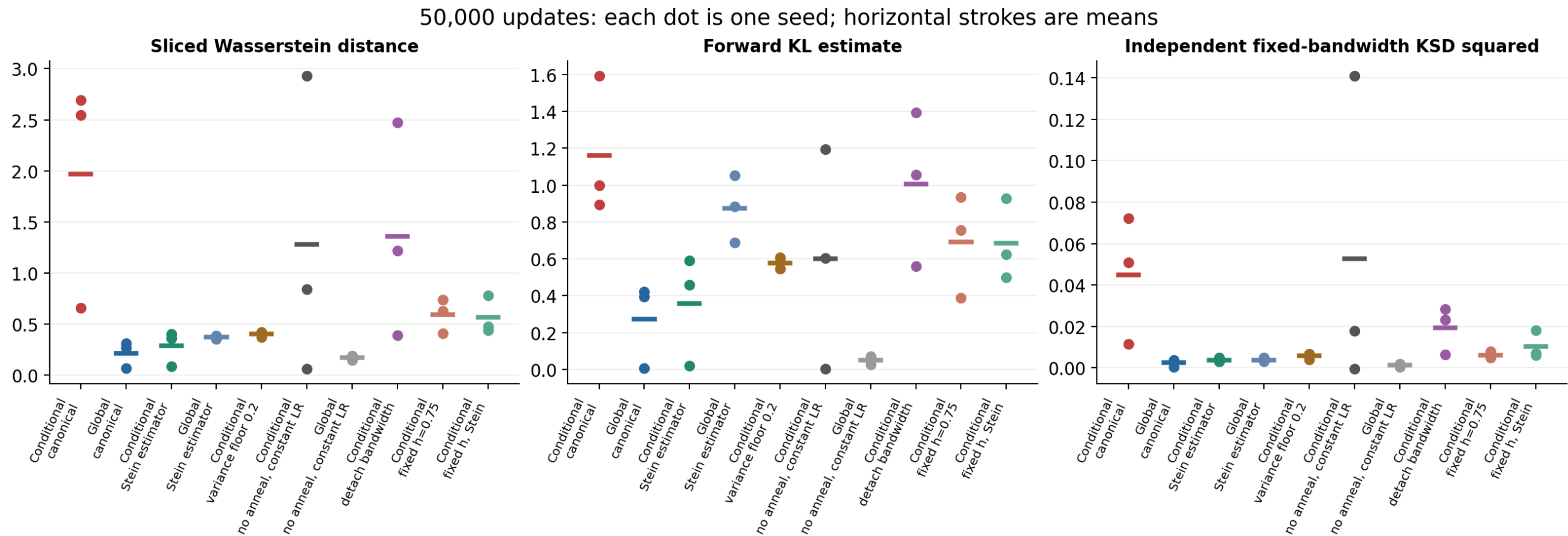
Cheng et al. (2024), Kernel Semi-Implicit Variational Inference, ICML / PMLR 235.
<https://proceedings.mlr.press/v235/cheng24l.html>

Yu et al. (2026), A Kernel Approach for Semi-implicit Variational Inference. <https://arxiv.org/abs/2601.12023>.
Appendix A.4 records global variance in their experiments.

Korba et al. (2021), Kernel Stein Discrepancy Descent, ICML / PMLR 139.
<https://proceedings.mlr.press/v139/korba21a.html>

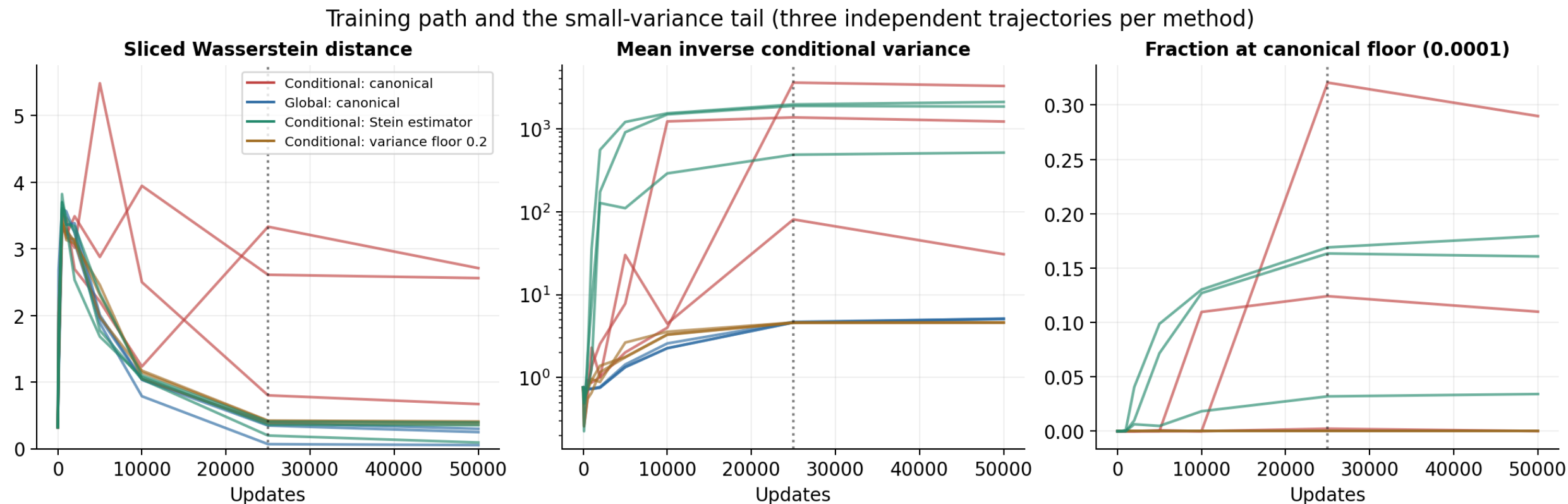
Original code: <https://github.com/longinYu/KSIVI>

Figure 1: confirmation



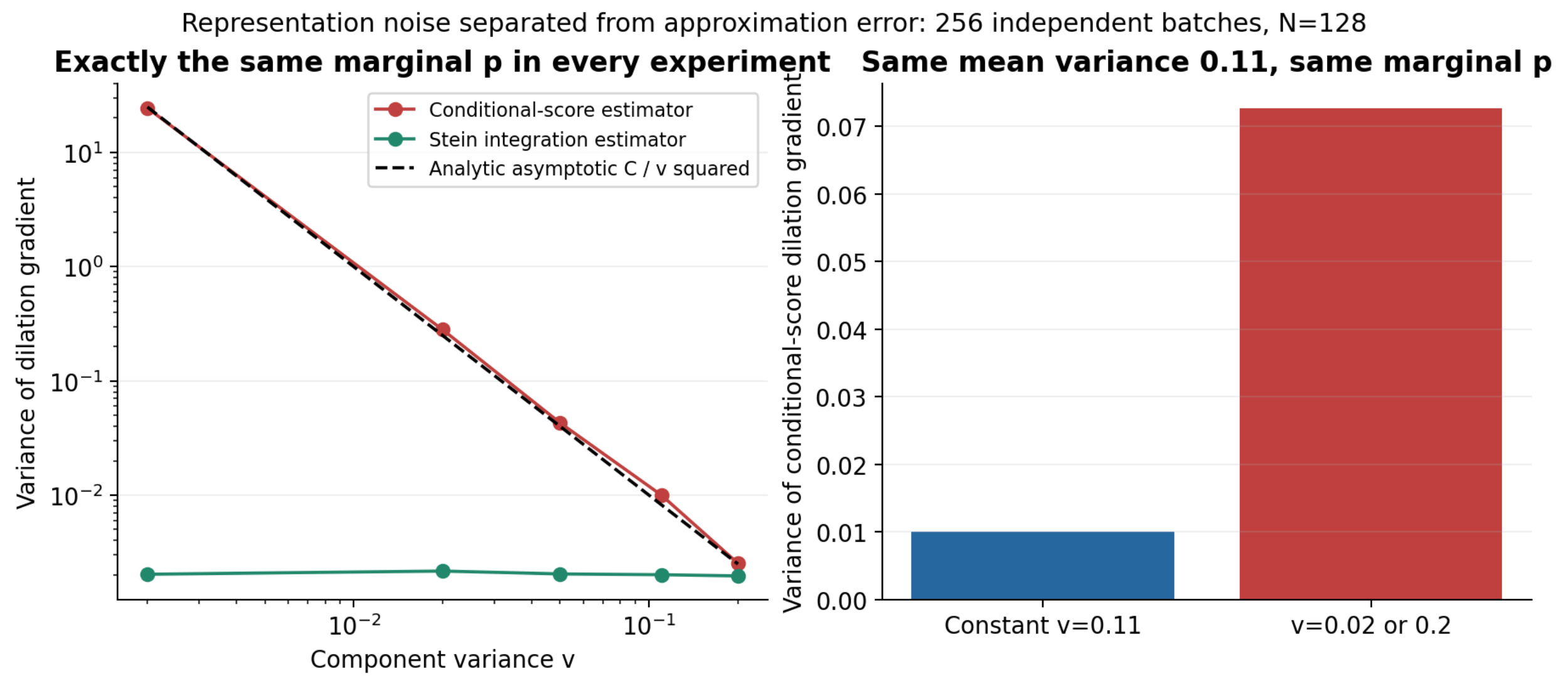
Matched initial mean/variance, seeds 42-44. KSD uses $h=0.75$ chosen independently of evaluation draws.

Figure 2: trajectories



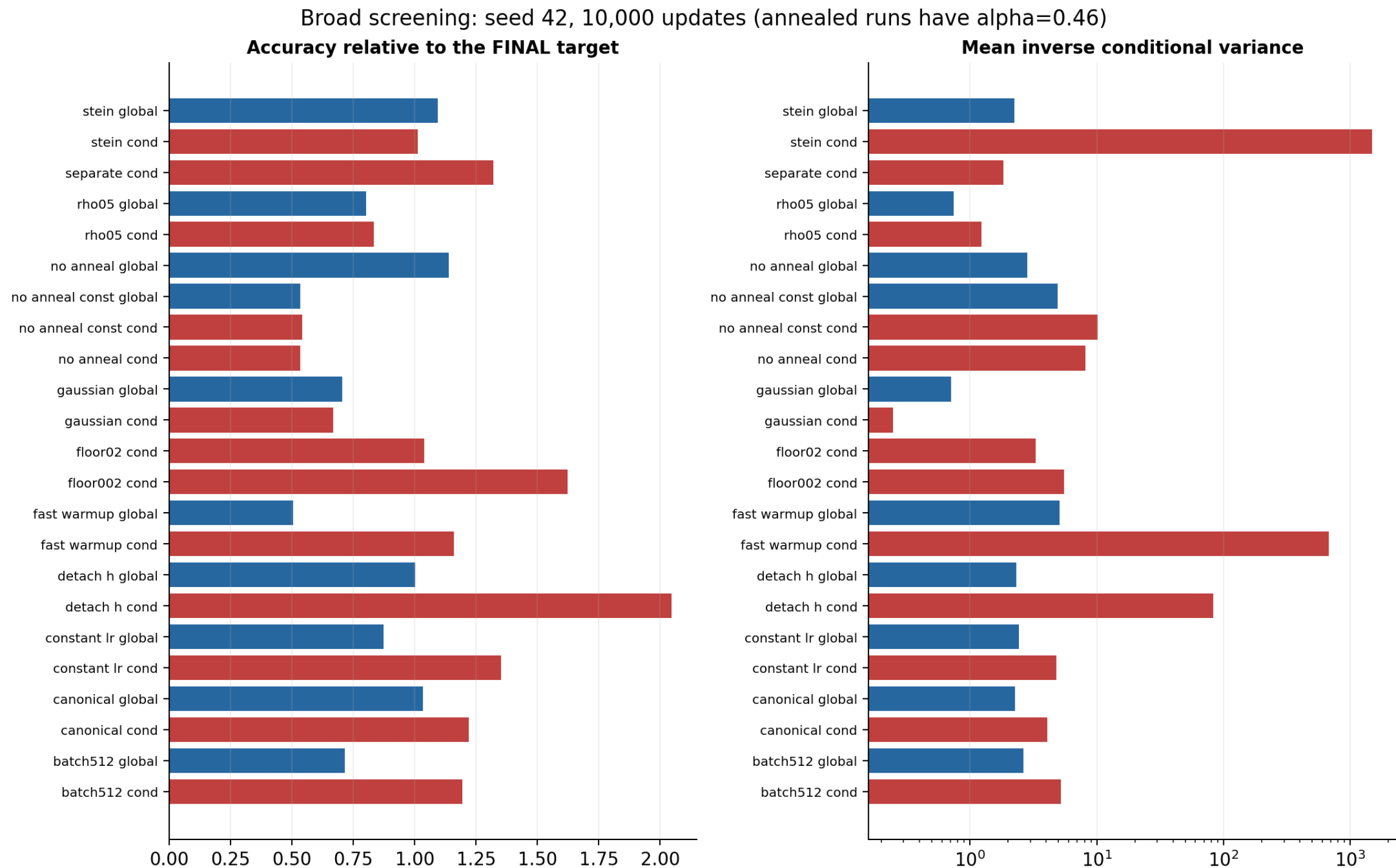
Vertical line: annealing reaches the final target. The floor fraction uses a common threshold of 0.00010001 across methods. Diagnostic RNG is isolated from training.

Figure 3: exact marginal noise



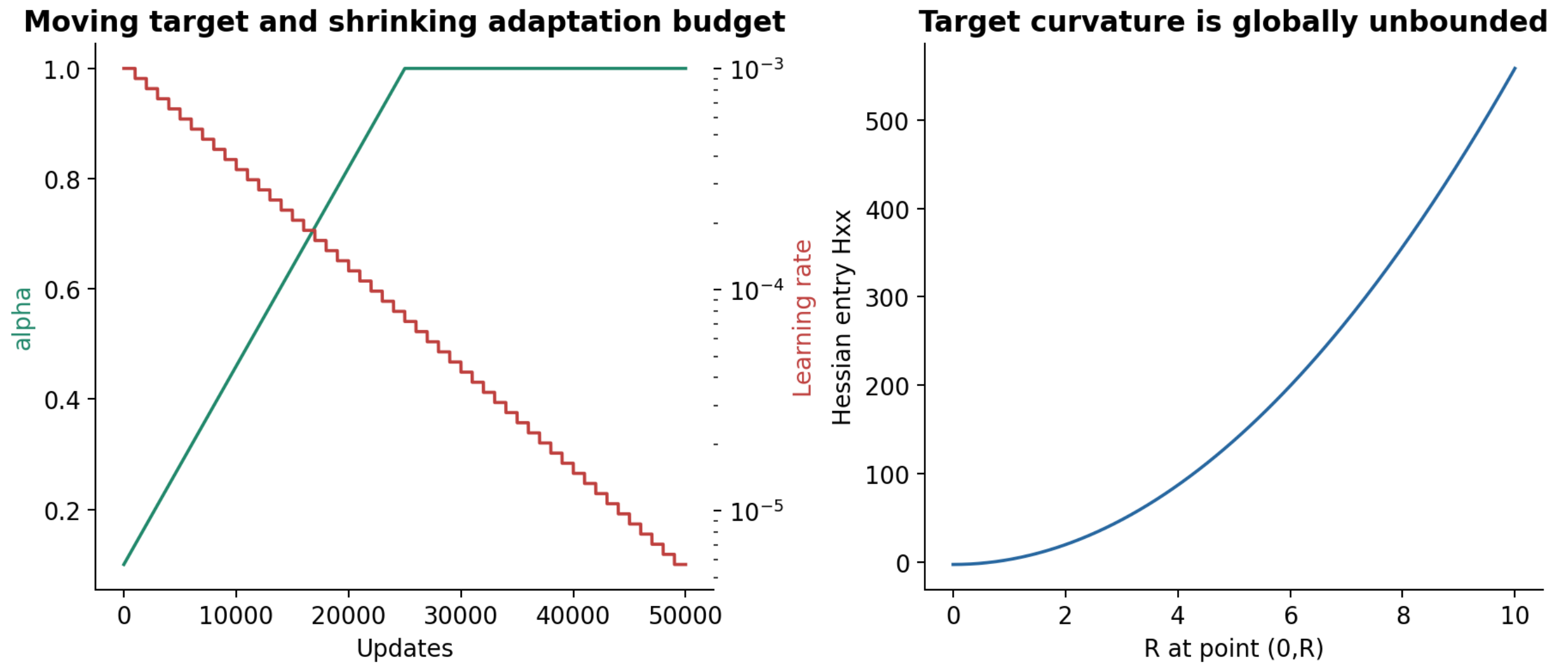
Global dilation parameter theta=0 is a population optimum. Fixed h=0.75; independent cross batches.

Figure 4: screening



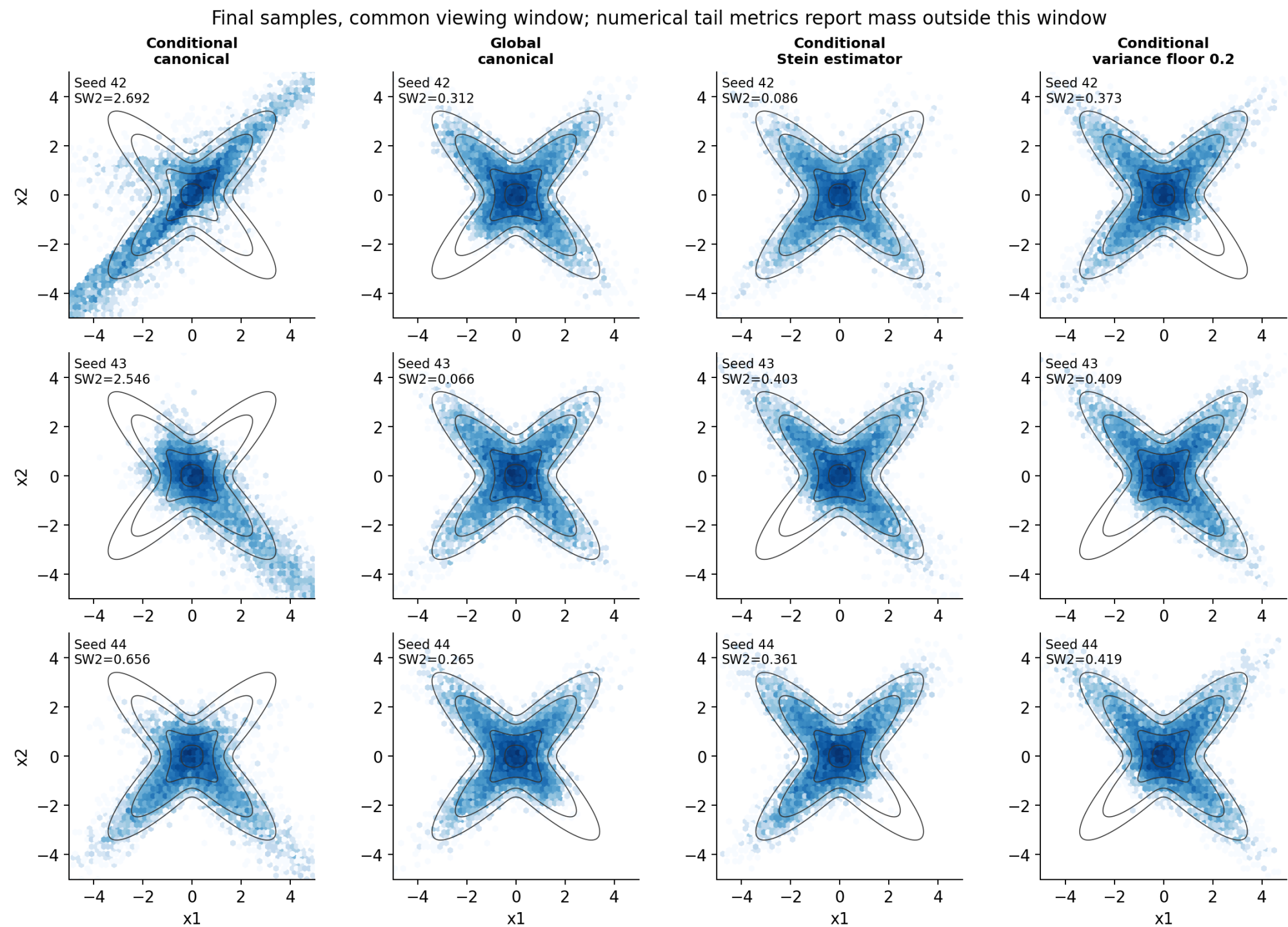
Screening ranks are not final-target convergence claims. Blue: global variance; red: conditional variance.

Figure 5: geometry and schedule



The X-shaped mixture violates the global bounded-Hessian assumption. This is a boundary of the theorem, not by itself a cause of the family gap.

Figure 6: final samples



Hexbin densities are for visual comparison. Quantitative metrics include all samples, including escaped tails.